The Color Beats Challenge

In this activity, we'll use MATLAB® to turn sounds into colors!

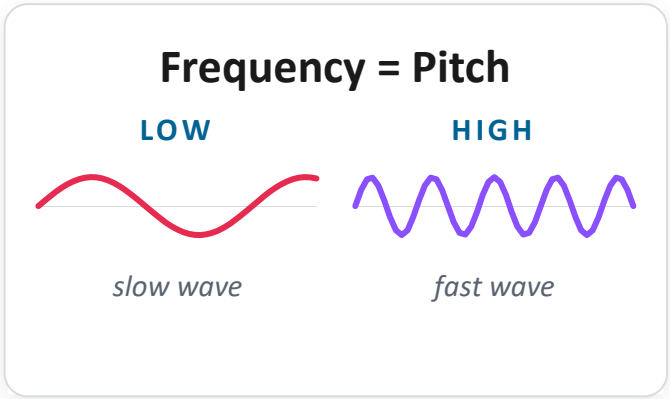
What color is your voice?

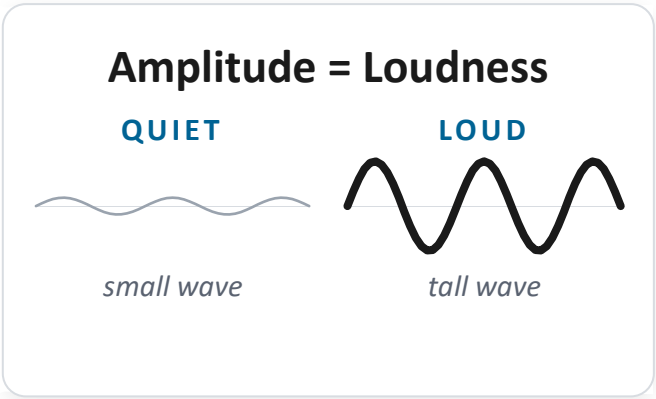


Background

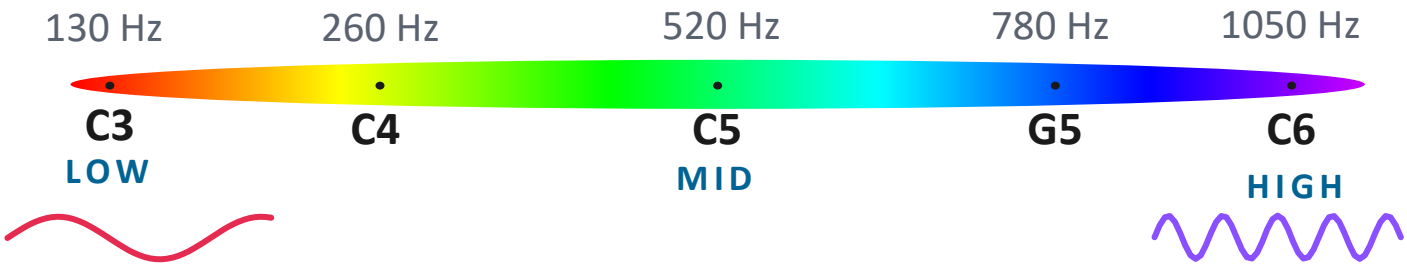
Sounds are waves

- The *pitch* is related to the wave *frequency* (how many waves are there?)
- The *loudness* is related to the wave *height* (how big is the wave?)





What colors do sounds make?



MathWorks

Activity Walkthrough

1 Set Recording

Choose how many seconds to record.

A quick hum? 3 seconds.

2 Press Record

The code counts down: 3, 2, 1, GO!

Talk, hum, sing, or whistle.

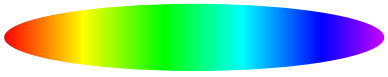
3 See your sound

Check out the pictures your sounds make!

In each Part of the Activity, you'll discover the different pictures you can use to represent sound:

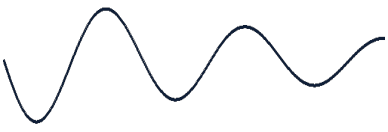
1 Color Key

Assign each pitch a color



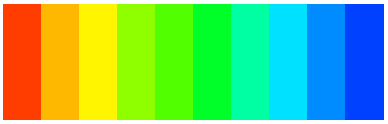
2 Waveform

Convert your sound into a wave



3 Stripes

Your wave broken up into strips

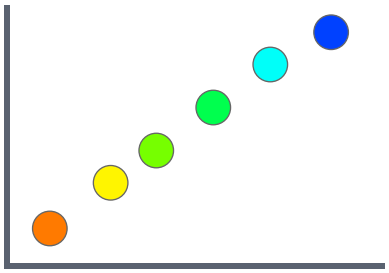


MathWorks

4

Scatter Plot

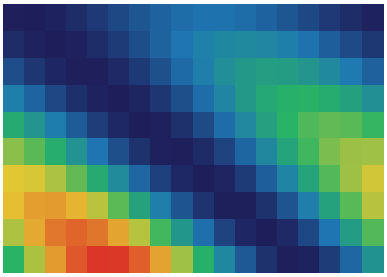
The exact pitch per second



5

Spectrogram

A painting of your sounds



Try These Challenges!

Make a rainbow

Slide your voice from your lowest hum to your highest whistle. Can you get red to violet in one breath?

Paint it black

Silence looks like black stripes. Can you make silence in the MIDDLE of your recording?

Draw a shape

Different sounds create different shapes in a spectrogram. Can you make a triangle? A house? A staircase?

Hit a target color

Someone picks a color. You find the pitch that paints it. Closest match wins.